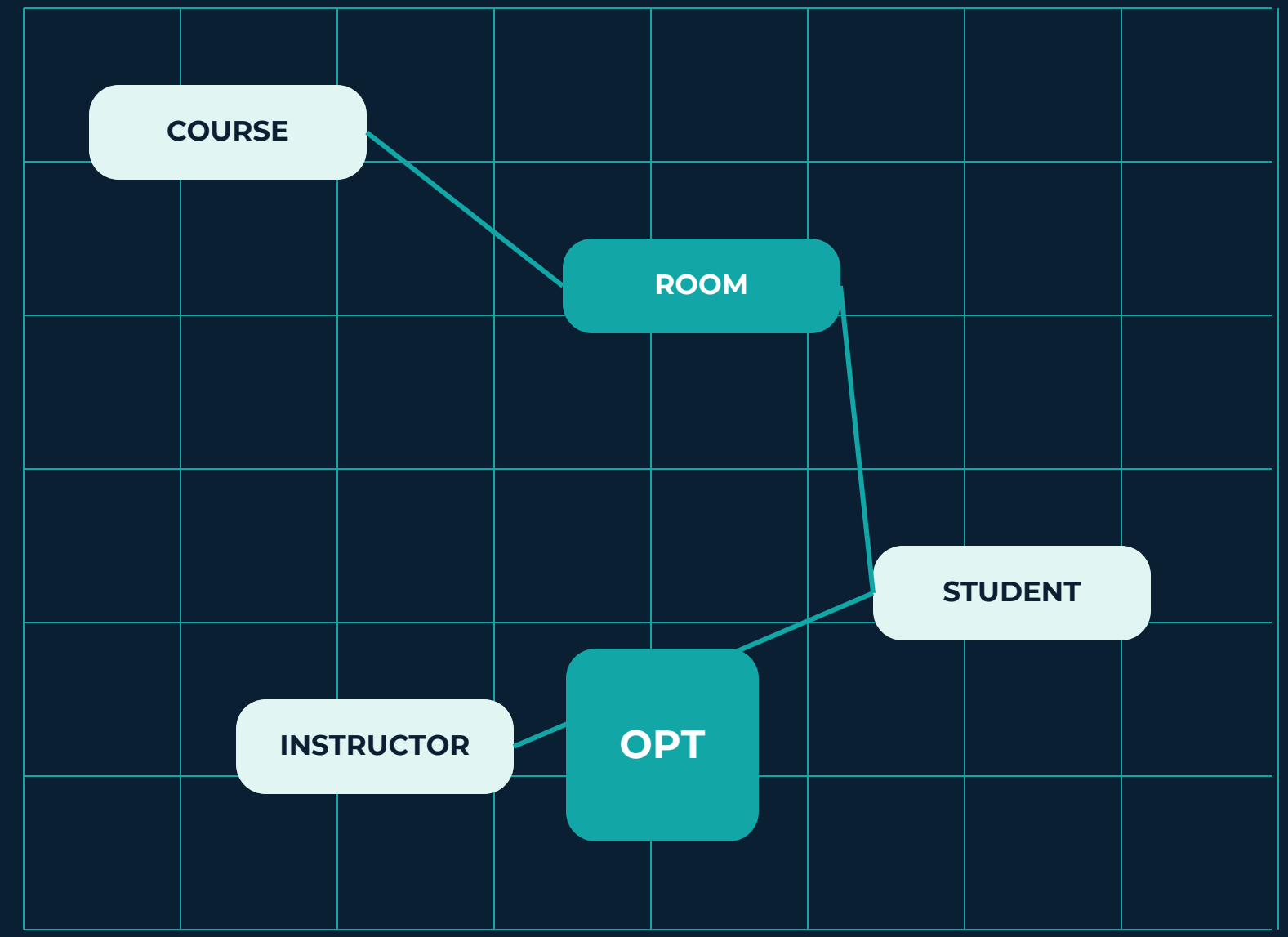


GRADUATION PROJECT PRESENTATION

Tanseek

Smart Timetable & Room Allocation System

Building conflict-free academic schedules through intelligent optimization.



Academic Scheduling Is More Complex Than It Looks



A single change can affect instructors, students, rooms, laboratories, and the entire timetable.

The Problem We Are Solving

01 Manual scheduling is time-consuming

02 Conflicts across departments

03 Instructor double-bookings

04 Room double assignments

05 Student section overlaps

06 Capacity and equipment issues

07 Availability is overlooked

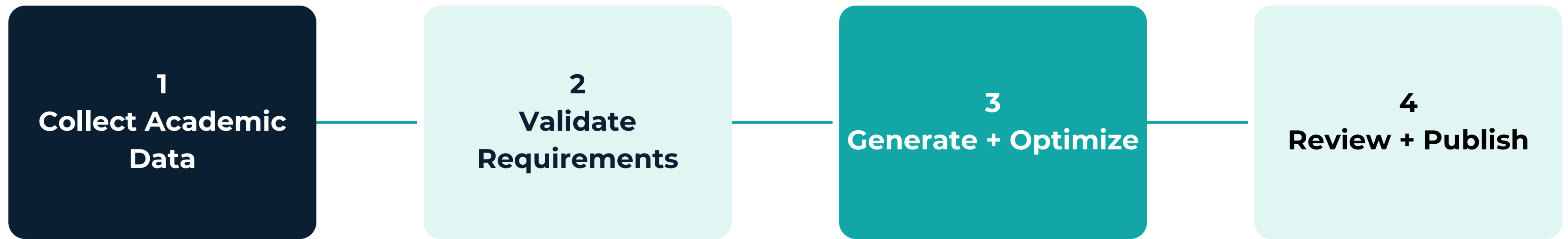
08 Changes create new conflicts

09 Failures are hard to explain

10 Timetables become outdated

“Creating a timetable is not only about selecting a time and a room. It is a complex constraint-satisfaction problem.”

Introducing Tanseek



ONE SYSTEM. ONE SOURCE OF TRUTH. ONE COORDINATED ACADEMIC SCHEDULE.

Tanseek turns validated academic data into a reviewable, explainable and publishable schedule.

Who Is Tanseek Built For?

SA

Super Admin

Accounts, roles, departments, permissions, audit logs

DC

Department Coordinator

Courses, sections, assignments and distributions

RO

Registration Officer

Student records and registrations

LT

Lecturer & TA

Availability and personal schedules

LM

Lab Manager

Rooms, equipment and capacity

SC

Scheduler

Generate drafts and review conflicts

AD

Admin

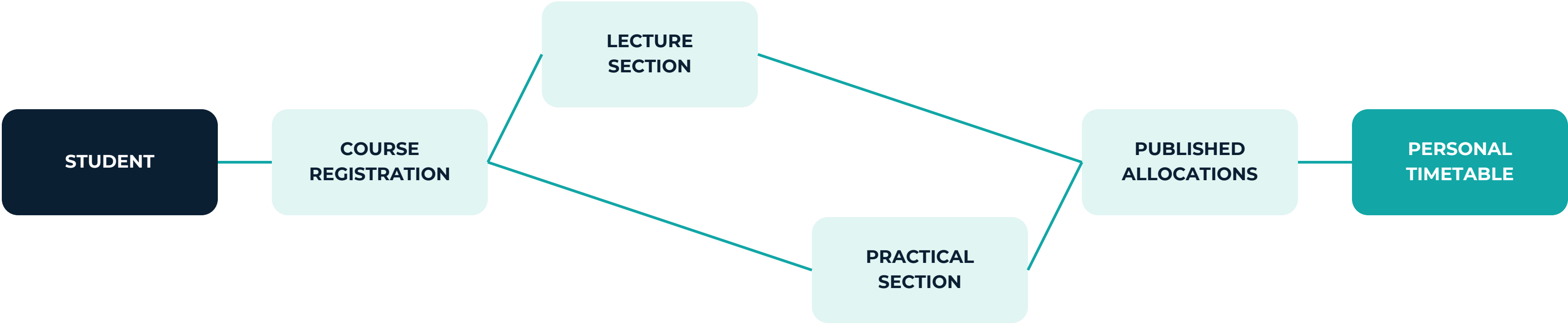
Edit and publish by department

ST

Student

Personal timetable and assignments

From Course Registration to a Personal Timetable



Every course can include Lecture and Practical sessions, each with multiple sections.

Student Groups support bulk organization — individual Section Enrollment remains authoritative.

What Data Does the System Use?

01

Academic Data

Terms, departments, courses, requirements, sections, durations

02

People

Students, registrations, enrollments, lecturers, TAs, availability

03

Resources

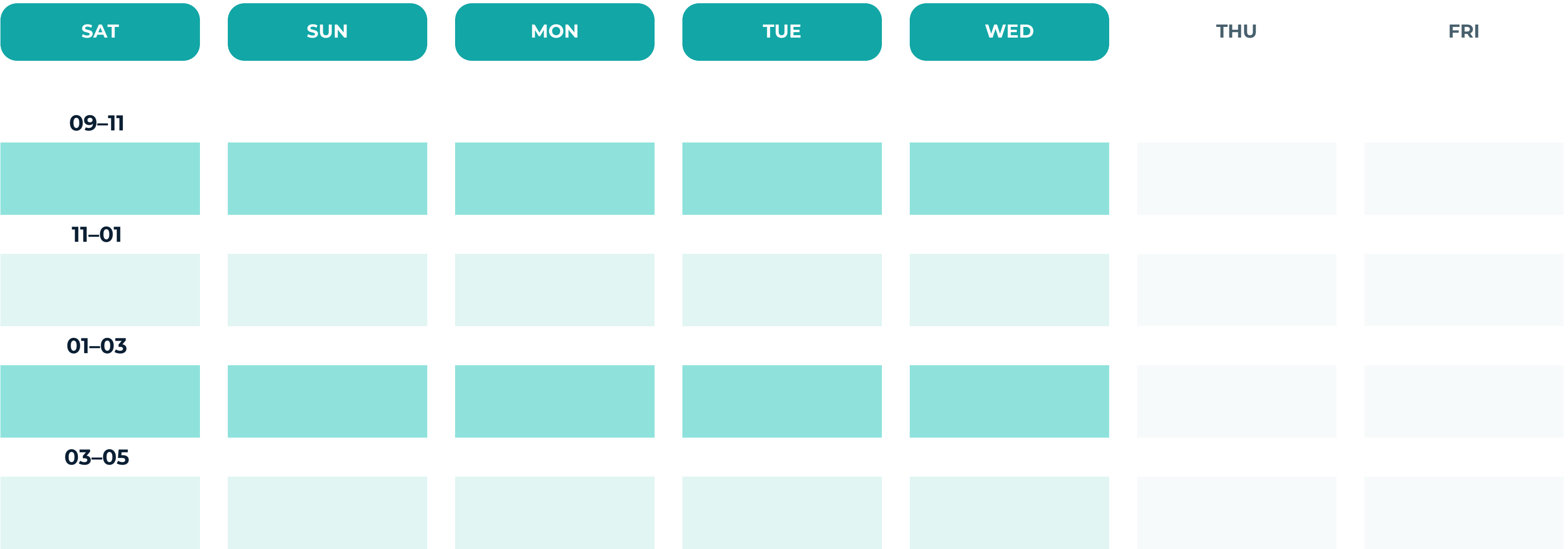
Classrooms, labs, capacity, equipment, closures

04

Scheduling

Study days, time slots, holidays, versions, allocations

Current Scheduling Policy



09:00 AM — 05:00 PM

2-hour slots • No fixed breaks

Availability deadline / AUG 31

How the Smart Scheduling Engine Works

HARD CONSTRAINTS

- ✓ Instructor conflict
- ✓ Student conflict
- ✓ Room conflict
- ✓ Availability
- ✓ Room type + capacity
- ✓ Equipment + eligibility
- ✓ Closures + holidays
- ✓ Valid assignments

SCHEDULING ENGINE

ONE OPTIMIZED
DRAFT SCHEDULE

SOFT CONSTRAINTS

- + Preferred instructor time
- + Best room fit
- + Equipment match
- + Compact schedules
- + Efficient utilization
- + Reduced gaps

Tanseek Does Not Only Detect Conflicts — It Explains Them

TANSEEK

Dashboard

Schedules

Conflicts

Versions

Conflict Review / Draft v03

INSTRUCTOR_CONFLICT

STUDENT_CONFLICT

ROOM_CONFLICT

CAPACITY_EXCEEDED

EQUIPMENT_SHORTAGE

NO_FEASIBLE_SLOT

RANKED ALTERNATIVES

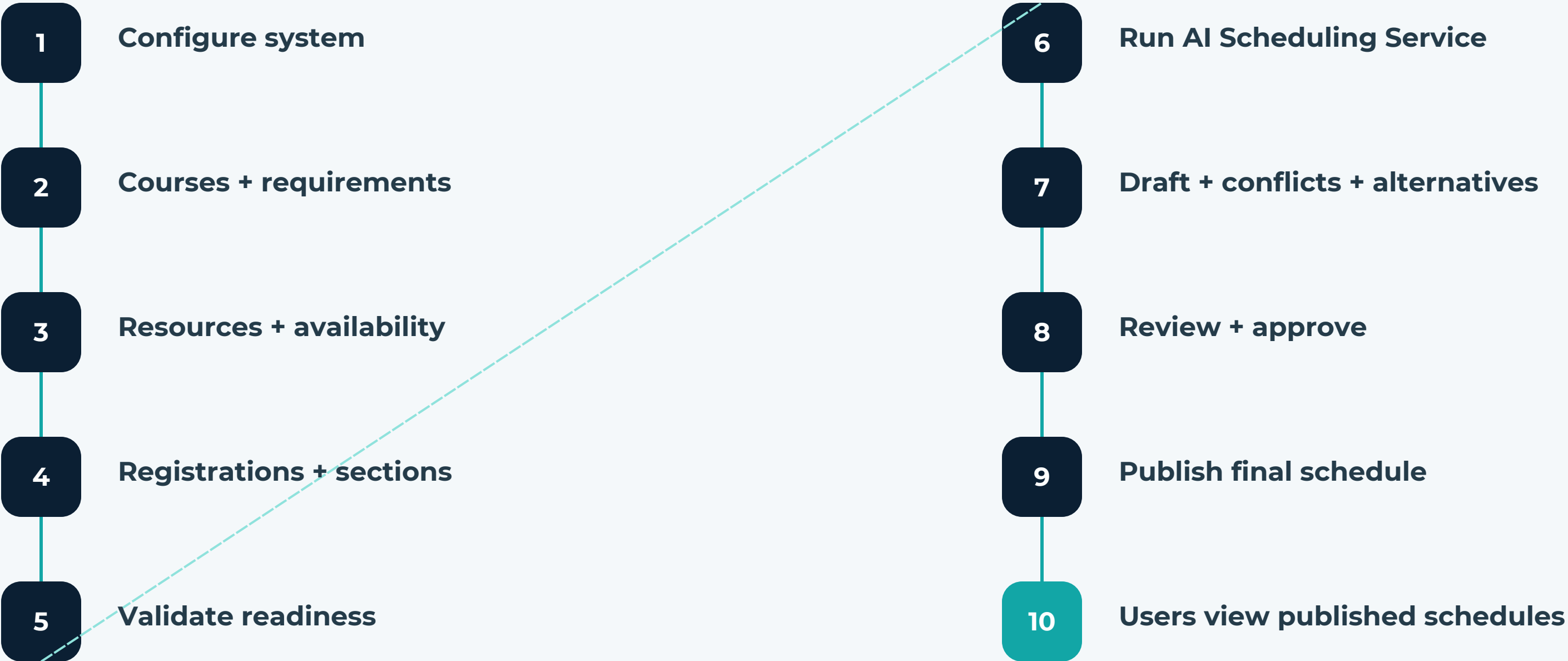
01 Move to Lab B-204 / 11:00

02 Keep room / shift to 13:00

03 Assign eligible instructor / 15:00

Exact reasons returned for unscheduled sessions with ranked feasible alternatives.

End-to-End Workflow



ONLY THE PUBLISHED VERSION IS VISIBLE TO END USERS

Core Features

01 Smart generation	02 Room + lab allocation	03 Student conflict detection	04 Instructor availability
05 Lecture + practical sections	06 Course registration	07 Section assignment	08 Capacity validation
09 Equipment matching	10 Schedule versioning	11 Draft review + publishing	12 Ranked alternatives
13 Conflict explanations	14 Role-based access	15 Department permissions	16 Audit logs
17 Personal timetables			

The Right Access for Every Role



Extra department access requires Super Admin approval.

Why Tanseek Matters

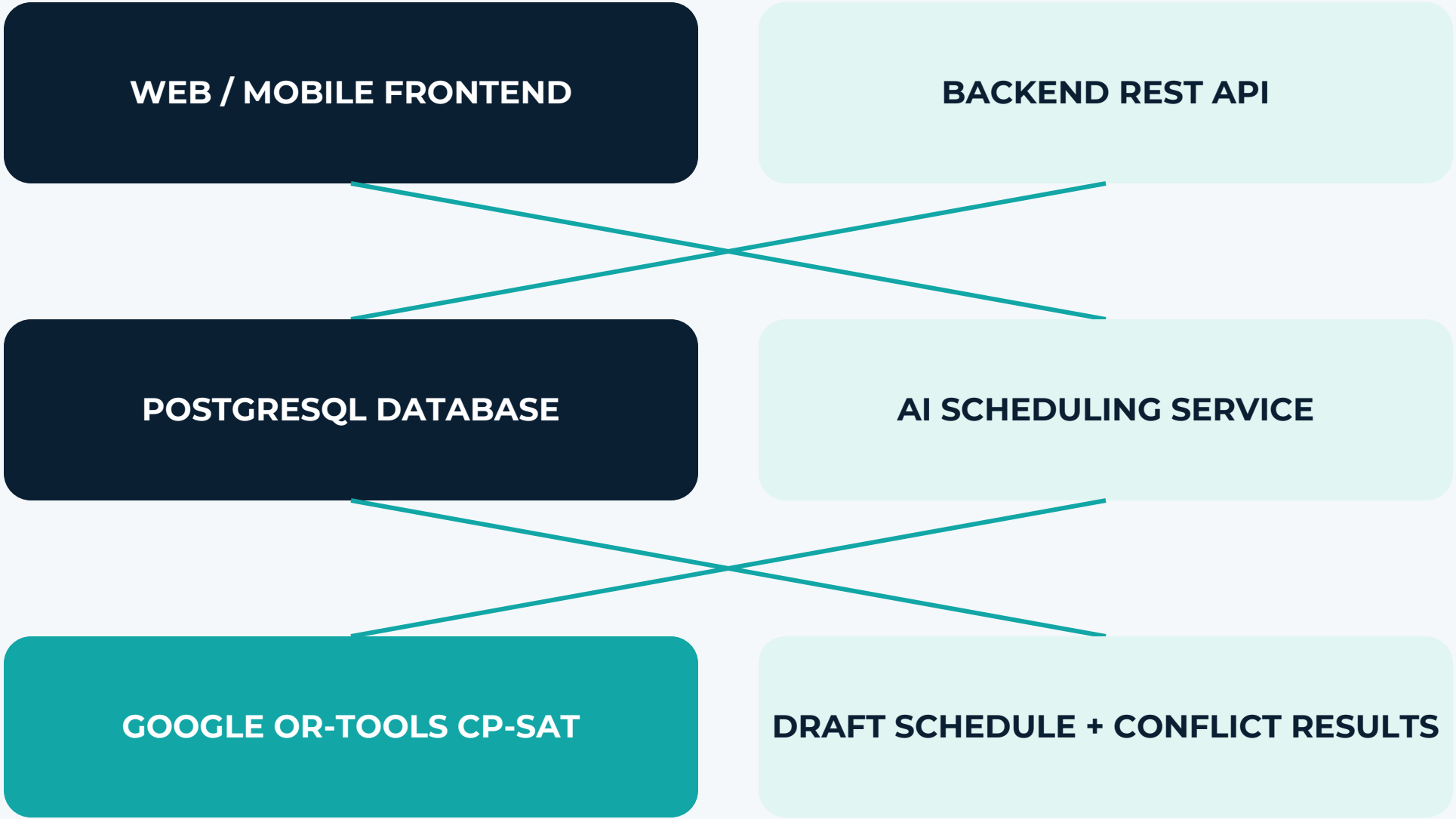
BEFORE TANSEEK

- Manual scheduling
- Repeated conflicts
- Scattered data
- Poor room utilization
- Difficult updates
- No explanations
- Multiple timetable copies

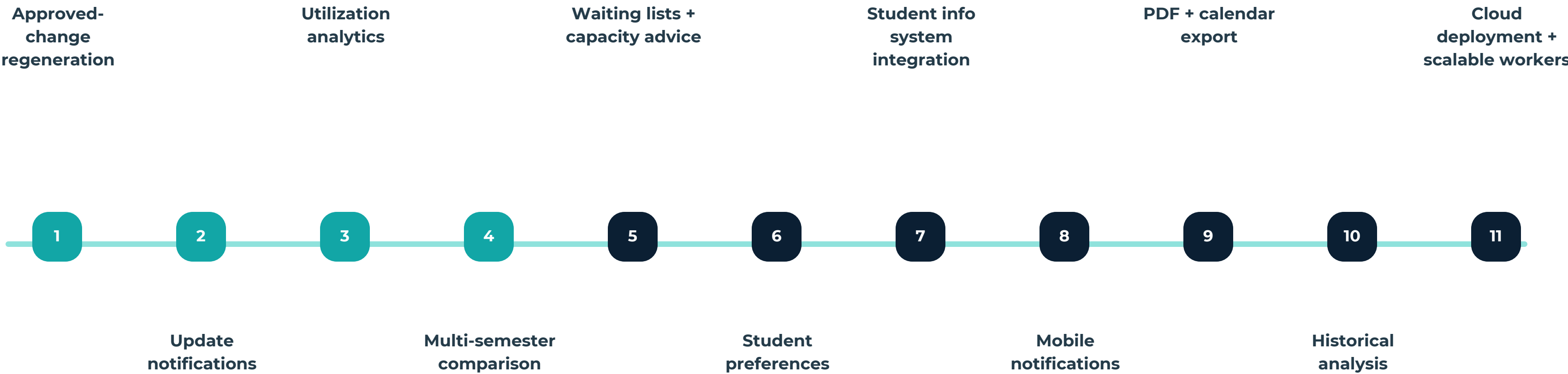
WITH TANSEEK

- ✓ **Automated generation**
- ✓ **Early conflict detection**
- ✓ **Centralized data**
- ✓ **Better room + lab use**
- ✓ **Faster revisions**
- ✓ **Explainable alternatives**
- ✓ **One published source of truth**
- ✓ **Personal schedules for all**

System Architecture



Future Enhancements



Meet the Team



**Beshoy Nashaat Nabil
Naguib**

Team Leader / Data &
Database Design



John Ayman Makram

DevOps



Mennatallah Mohsen

AI Engineering / Scheduling
Model



Omar Mohamed

Frontend / Mobile
Development



Osama Ramzy

Backend Development



From complex academic constraints to one clear, optimized schedule.

Smart Scheduling. Better Resources. Fewer Conflicts.

THANK YOU / QUESTIONS?

